General Relativity: From the Equivalence Principle to Tensors

Based on lectures by Leonard Susskind

Lecture notes organized by LazyingArt LLC with Video2Book

Original lecture by Leonard Susskind. Notes reconstructed from the lecture and organized by LazyingArt LLC with Video2Book.

Chapter 1

From the Equivalence Principle to Tensors

General relativity is commonly shortened to GR, although special relativity is rarely called SR. The abbreviation is less important than the decision to begin at the beginning. The most direct route is the route Einstein took: start from a physical observation simple enough to be felt in an elevator, press it until its consequences become unavoidable, and only then introduce the geometry.

The first observation is the equivalence principle. Gravity and acceleration are locally indistinguishable. The word locally will eventually carry nearly the entire burden of the statement. Before qualifying it, however, we should see what the unqualified statement can already do.

1.1 The Elevator as a Coordinate Transformation

Textbooks often replace Einstein's elevator by a rocket ship. An elevator is preferable here because its acceleration is familiar. Let z measure height in a frame attached to the ground, and let z' measure height from the elevator floor. If the floor has height $L(t)$, the same event has coordinates

$$z' = z - L(t), \qquad t' = t, \qquad x' = x. \tag{1.1}$$

The horizontal coordinate is unchanged because the elevator does not slide sideways. Taking $t' = t$ deliberately suppresses special relativity for the moment.

Question from the audience. How can a course on general relativity begin by ignoring special relativity?

Response. There is a regime in which gravity matters while all velocities remain small compared with c. The gravity–acceleration connection can therefore be isolated in Newtonian mechanics and then combined with Minkowski geometry. This is a staging choice, not a claim that the final theory is nonrelativistic. [a]

[a] Lecture timestamp: 00:05:33.

For two inertial frames the elevator moves uniformly,

$$L(t) = vt, \qquad z' = z - vt. \tag{1.2}$$

The origins have been chosen to coincide at $t = 0$. The board keeps the elevator sketch beside all three coordinate relations so that the symbols never lose their physical meaning.

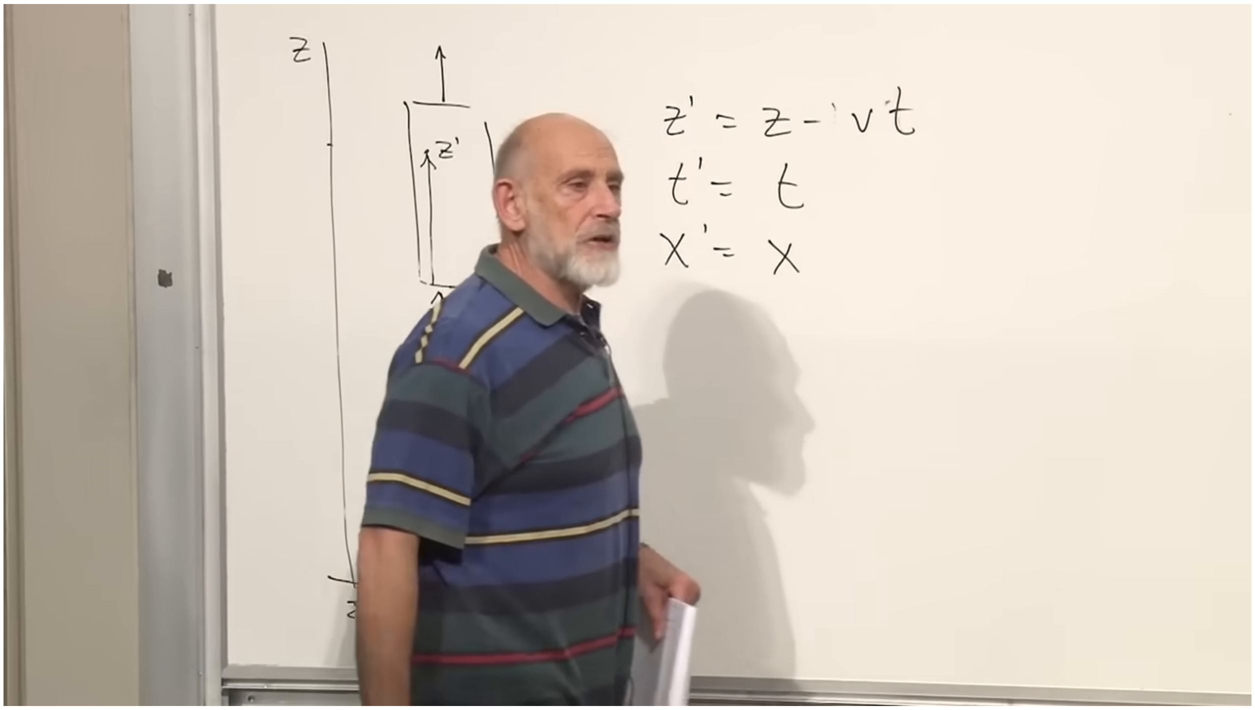

Figure 1.1: The elevator coordinates after the uniform-velocity specialization $L(t) = vt$.
Lecture frame: 00:09:10.

Consider only the vertical component of Newton's law,

$$F = m\ddot{z}. \tag{1.3}$$

The force may come from a rope, an electric field, or any other ordinary interaction. Differentiating (1.2) twice gives $\ddot{z}' = \ddot{z}$, and hence

$$F = m\ddot{z}'. \tag{1.4}$$

The form of the law is unchanged. This elementary calculation exposes the operative content of inertial-frame equivalence: a linear coordinate transformation takes uniform motion to uniform motion.

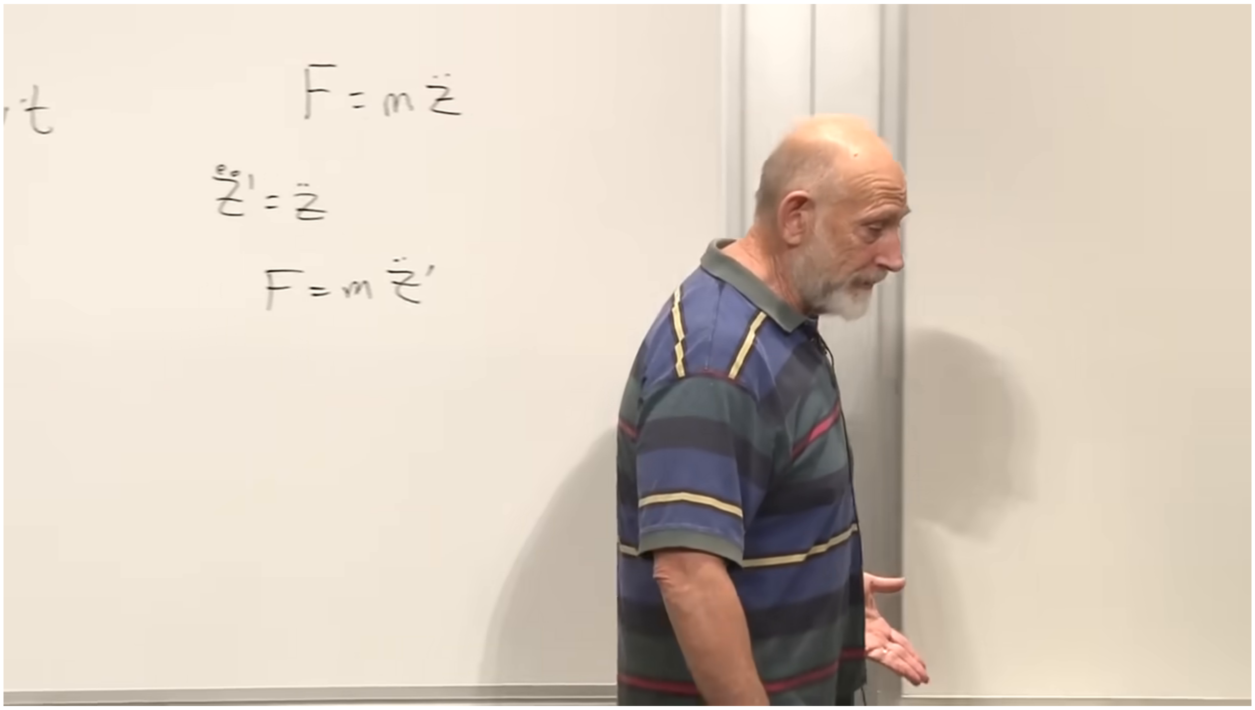

Figure 1.2: Newton's equation in the two uniformly moving coordinate frames.
Lecture frame: 00:11:50.

Now accelerate the elevator upward at constant g:

$$L(t) = \frac{1}{2}gt^2, \qquad \dot{L} = gt, \qquad \ddot{L} = g. \tag{1.5}$$

The coordinate transformation becomes

$$z' = z - \frac{1}{2}gt^2, \qquad \ddot{z}' = \ddot{z} - g. \tag{1.6}$$

Substitution into Newton's law gives

$$F = m(\ddot{z}' + g), \qquad m\ddot{z}' = F - mg. \tag{1.7}$$

The derivation includes a useful live sign check: the $+mg$ first appears on the acceleration side and becomes $-mg$ only when moved to the force side. The final direction is downward in the frame whose floor accelerates upward.

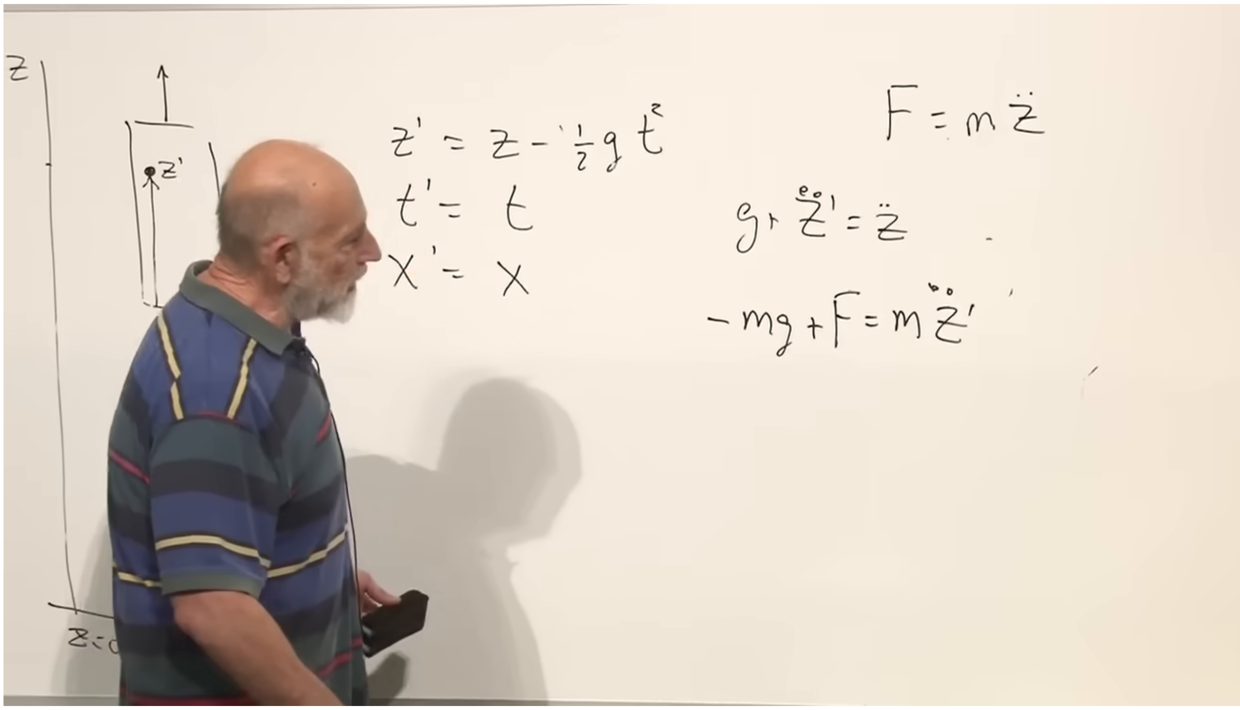

Figure 1.3: The accelerated transformation and the resulting fictitious force $-mg$.
Lecture frame: 00:15:35.

The new term looks gravitational for a precise reason: it is proportional to mass. Since inertial response is also proportional to mass, the mass cancels from the equation of motion. A freely falling body follows the same trajectory regardless of its mass. The orbit of the Earth, for example, is fixed by its position and velocity in the Sun's field, not by the Earth's own mass in the test-body approximation.

Before Einstein, the resemblance between accelerated-frame forces and gravity could be treated as a curiosity. Einstein promoted it to a principle: no local mechanical experiment should distinguish a homogeneous gravitational field from an appropriately accelerated frame.

1.2 Straight Lines, Curved Coordinates, and Light

Plot t vertically and z horizontally. For $z' = z - vt$, lines of constant z' are parallel straight lines. Straight worldlines remain straight because the coordinate transformation is linear.

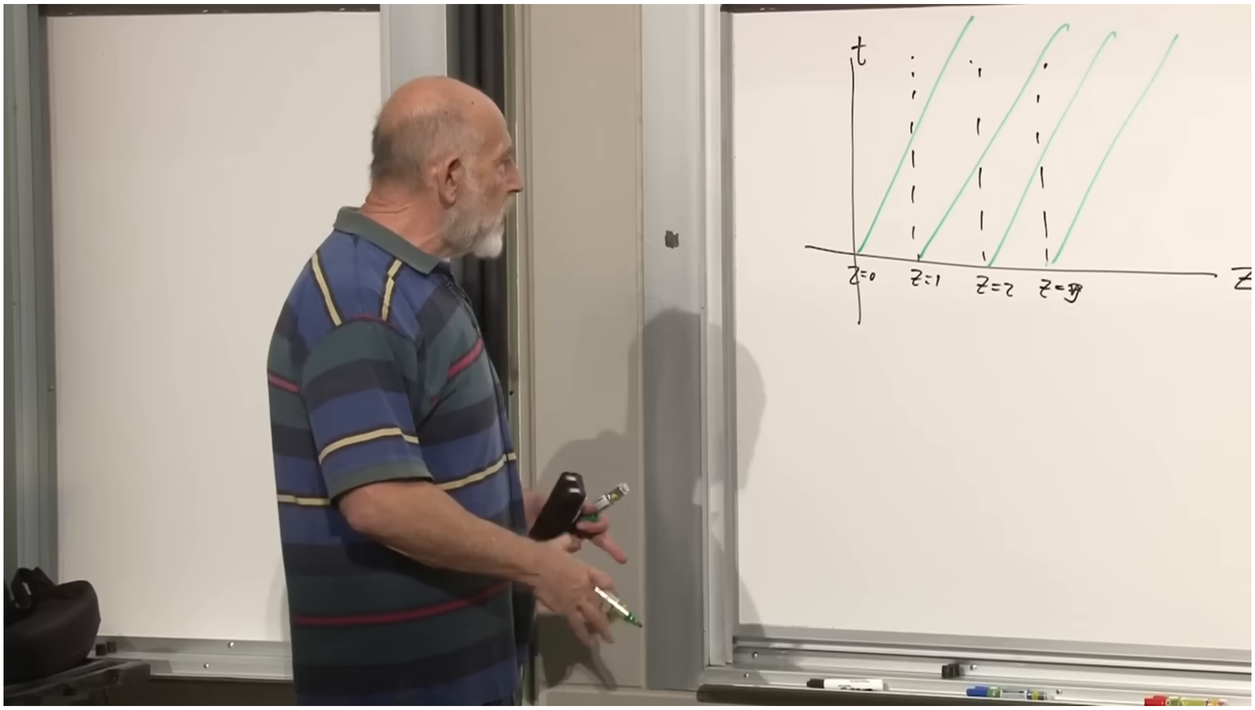

Figure 1.4: A linear spacetime grid for uniformly moving coordinates.
Lecture frame: 00:19:45.

For the accelerated map, a line of constant z' satisfies

$$z = z' + \frac{1}{2}gt^2. \tag{1.8}$$

It is a parabola lying on its side. Extending the motion backward in time gives an elevator that descends while slowing, turns, and then accelerates upward. Every neighboring constant-z' line is the same parabola shifted horizontally.

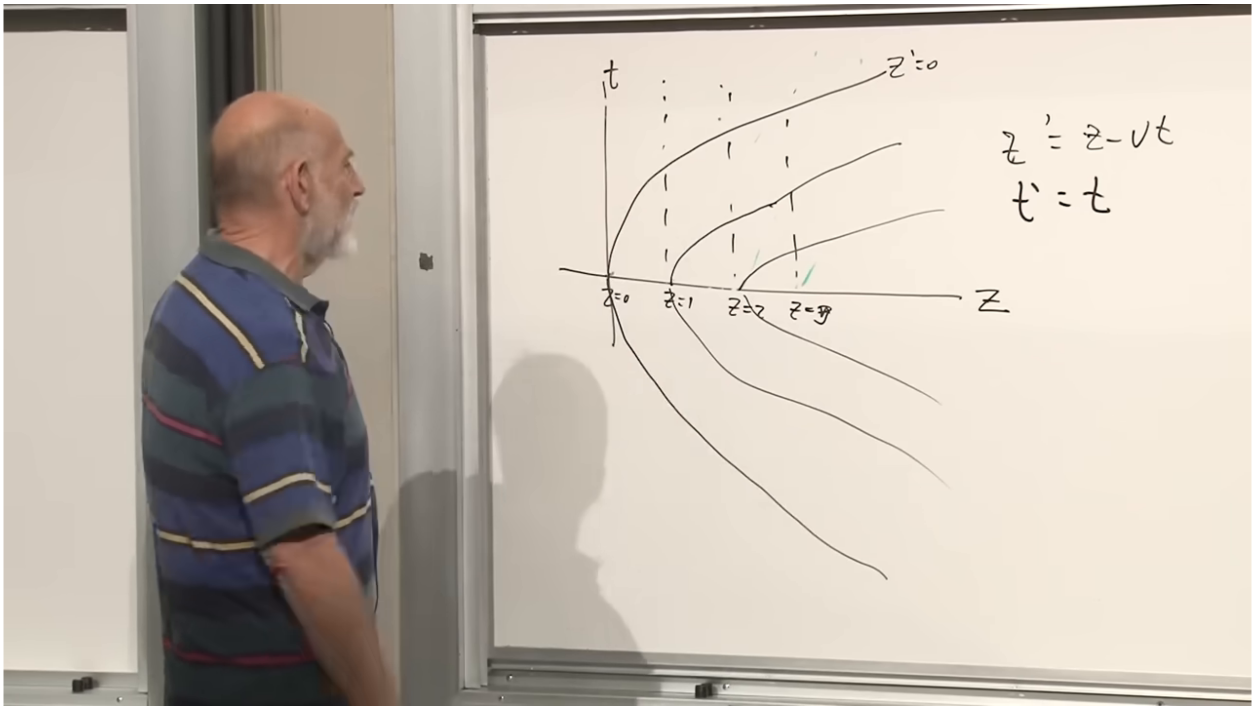

Figure 1.5: Constant-z' curves for the uniformly accelerated coordinate system.
Lecture frame: 00:22:25.

This is the first geometric clue. Special relativity uses linear Lorentz transformations, which carry straight spacetime lines to straight spacetime lines. To represent acceleration, and therefore to imitate gravity, one needs curvilinear transformations of spacetime coordinates.

Einstein used this apparently modest point to answer a question that was not modest at all: does gravity affect light? At $t = 0$, let the elevator and ground frames be instantaneously at rest relative to one another, and send a horizontal light ray across the elevator. In the unaccelerated frame,

$$x = ct, \qquad z = 0. \tag{1.9}$$

Using $z = z' + \frac{1}{2}gt^2$, the same ray in elevator coordinates obeys

$$z' = -\frac{1}{2}gt^2 = -\frac{g}{2c^2}x^2. \tag{1.10}$$

The light ray is straight in the inertial frame and curved in the accelerated frame. If acceleration and gravity are locally equivalent, a gravitational field must bend light. The argument establishes the effect before a field equation for gravity has been written.

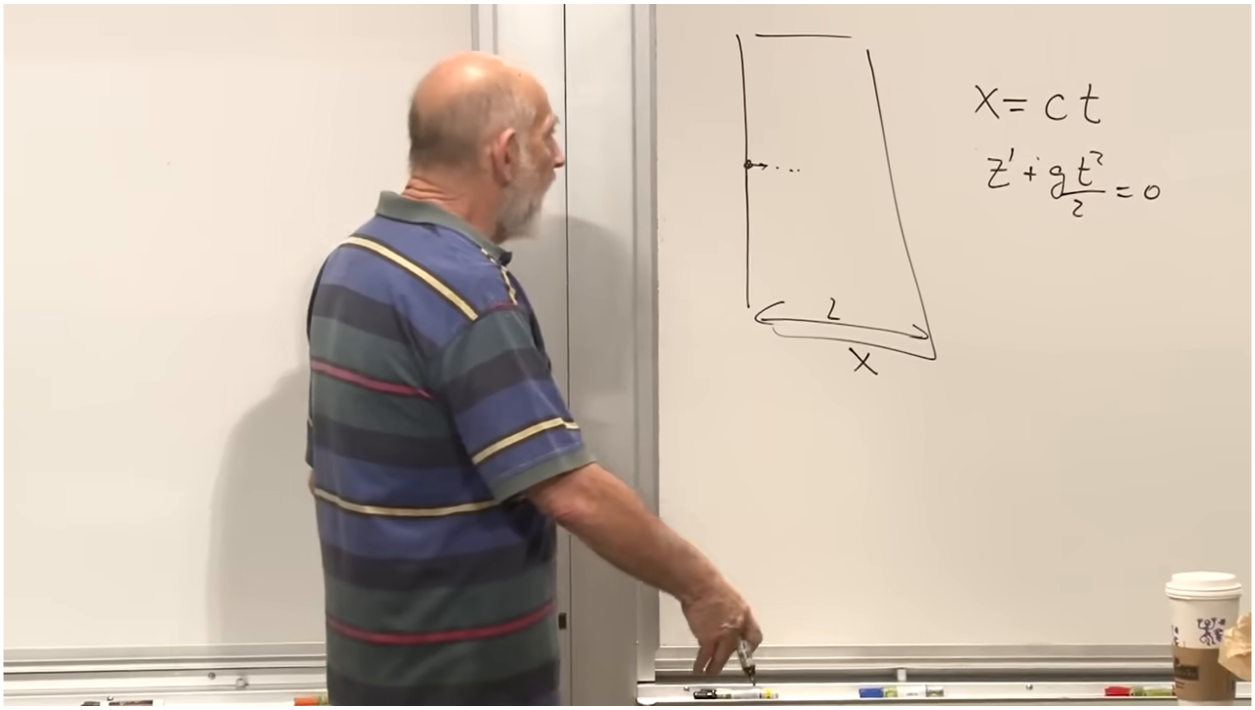

Figure 1.6: The horizontal light ray and its curved trajectory in the accelerated frame.
Lecture frame: 00:28:10.

1.3 What Cannot Be Transformed Away

An accelerated frame can manufacture an apparent uniform gravitational field. It does not follow that every gravitational field is a coordinate artifact. The field of a star points radially inward. A small freely falling laboratory near one point can erase the local acceleration, but the same transformation cannot erase the field on all sides of the star.

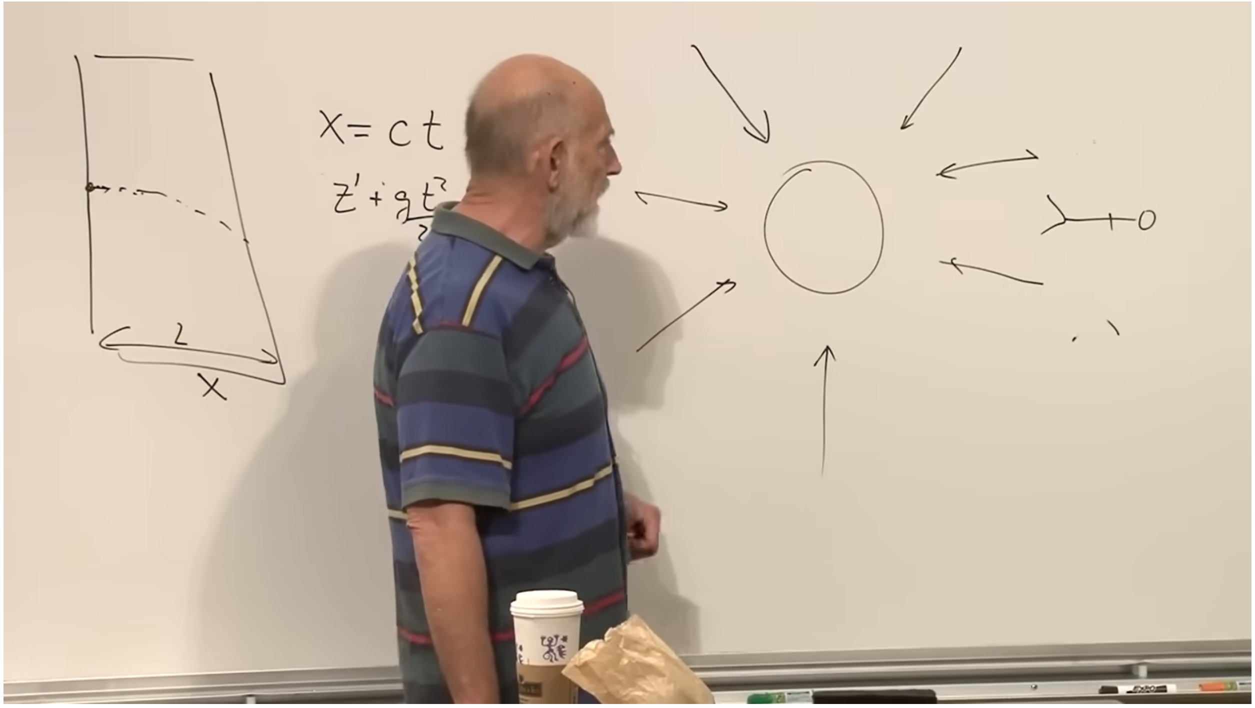

Figure 1.7: A radial field and an extended falling body used to expose tidal acceleration.
Lecture frame: 00:32:50.

The obstruction is tidal force. Imagine a two-thousand-mile person falling feet first. His feet are closer to the source and accelerate more strongly than his head, so he is stretched. Turn him sideways and the inward directions on opposite sides have components toward one another, so he is squeezed. In Newtonian language, neighboring points separated by ξ^j acquire relative acceleration

$$\delta a^i \simeq \partial_j g^i \, \xi^j. \tag{1.11}$$

A coordinate acceleration can be removed. Relative acceleration across an extended freely falling body cannot.

Question from the audience. Could different rocket engines at the head and feet simulate the stretching and therefore simulate a tidal field?

Response. Applied engines can certainly stress a body, but that is a real non-gravitational force acting differently on its parts. A coordinate change alone cannot create invariant relative acceleration among neighboring freely falling particles. The diagnostic concerns unforced free fall, not a body driven by separate rockets. [a]

[a] Lecture timestamp: 00:36:51.

Curved coordinates can nevertheless make fictitious fields look quite complicated. A frame may accelerate vertically while oscillating horizontally, producing a time-dependent apparent field with both components. Even ordinary polar coordinates provide an instructive example. A free particle moving on a straight Cartesian line first has decreasing $r(t)$, reaches a minimum radius, and then has increasing $r(t)$. Someone given only $r(t)$ and $\theta(t)$ might infer an outward repulsive force. In rotating variants of these coordinates, centrifugal and Coriolis terms appear. They are proportional to mass and therefore look gravitational, but they leave no invariant tidal strain on a freely falling body.

Question from the audience. What happens if the motion is described in angular or polar coordinates?

Response. The nonlinear coordinate description makes a straight free trajectory look as though it is repelled from $r = 0$. That effective centrifugal force is a coordinate force. Returning to an inertial Cartesian frame removes it and reveals no tidal distortion. [a]

[a]Lecture timestamp: 00:37:56.

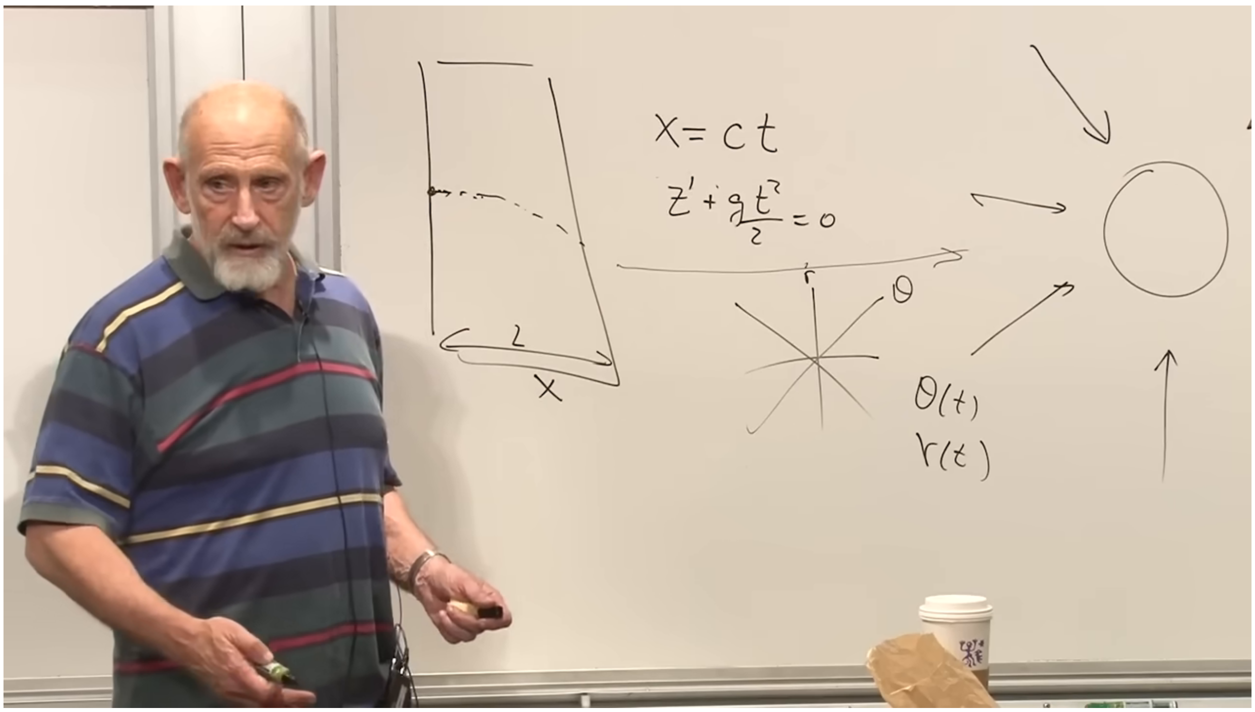

Figure 1.8: The polar-coordinate description of a straight trajectory, with $r(t)$ and $\theta(t)$ marked on the board.

Lecture frame: 00:40:20.

Question from the audience. Could several gravitating bodies cancel one another and leave no tidal force?

Response. Forces can cancel at a chosen point, but cancellation of the field value does not generally cancel its spatial derivatives. Near any one source the neighboring trajectories still acquire different accelerations. A true absence of curvature requires the tidal diagnostic to vanish throughout the region, not merely the net force at one point. [a]

[a]Lecture timestamp: 00:42:16.

Question from the audience. How is a tidal force measured operationally?

Response. Release an extended object, such as a crystal, in free fall and look for stresses and strains that stretch it in one direction and compress it in another. Those deformations are measurable and do not disappear when coordinates are renamed. [a]

[a]Lecture timestamp: 00:44:38.

Question from the audience. Would a two-thousand-mile person on a merry-go-round feel a radial tidal effect?

Response. A person strapped to the carousel feels supporting forces and may indeed get a headache. The test object must instead be freely falling. Its trajectory looks complicated in rotating coordinates but remains an unstressed straight trajectory in the inertial frame, so the invariant tidal force is zero. [a]

[a]Lecture timestamp: 00:45:07.

Question from the audience. Can the falling body be tilted so that stretching and squeezing cancel?

Response. No single orientation removes the full deformation. The radial field stretches along one principal direction and compresses along transverse directions. A body with finite extent in all directions experiences the complete pattern. [a]

[a]Lecture timestamp: 00:47:10.

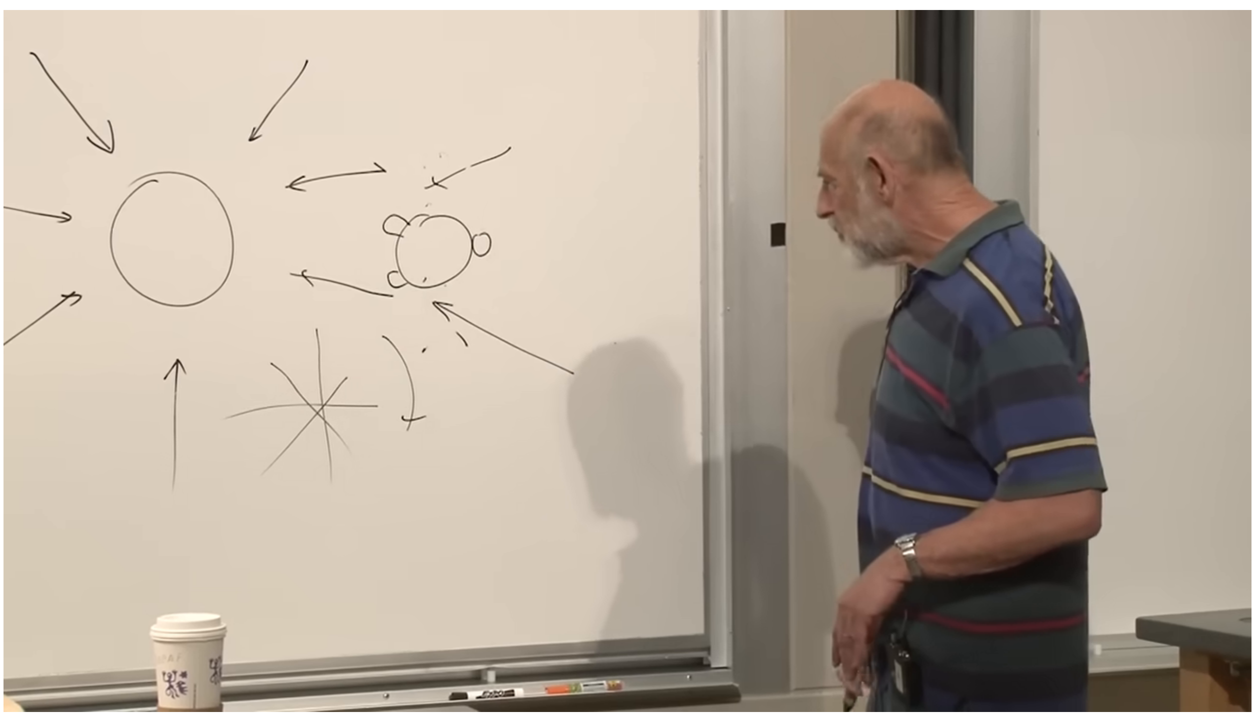

Figure 1.9: Radial stretching and transverse squeezing displayed together for an extended freely falling body.

Lecture frame: 00:47:55.

Question from the audience. Do tidal forces mean that the equivalence principle fails?

Response. They determine how carefully it must be stated. In a system small enough that the gravitational field varies negligibly across it, gravity can be removed to the relevant accuracy by a freely falling frame. The residual variation across a larger system is the tidal field. [a]

[a]Lecture timestamp: 00:48:08.

1.4 From Minkowski Geometry to a Local Metric

We now take a temporary rest from gravity. Minkowski had already taught that spacetime possesses an invariant local geometry. In one spatial dimension, with $c = 1$,

$$d\tau^2 = dt^2 - dx^2. \tag{1.12}$$

One may instead define $ds^2 = dx^2 - dt^2 = -d\tau^2$. The difference is a signature convention, not a difference in physics.

Question from the audience. Where does the speed of light belong if it is not set to one?

Response. If t is measured in time units and x in length units, one convenient form is $d\tau^2 = dt^2 - dx^2/c^2$. Equivalently, multiply by c^2 and use $c^2 d\tau^2 = c^2 dt^2 - dx^2$. Choosing ct or x/c as a coordinate absorbs the conversion factor. [a]

[a]Lecture timestamp: 00:51:02.

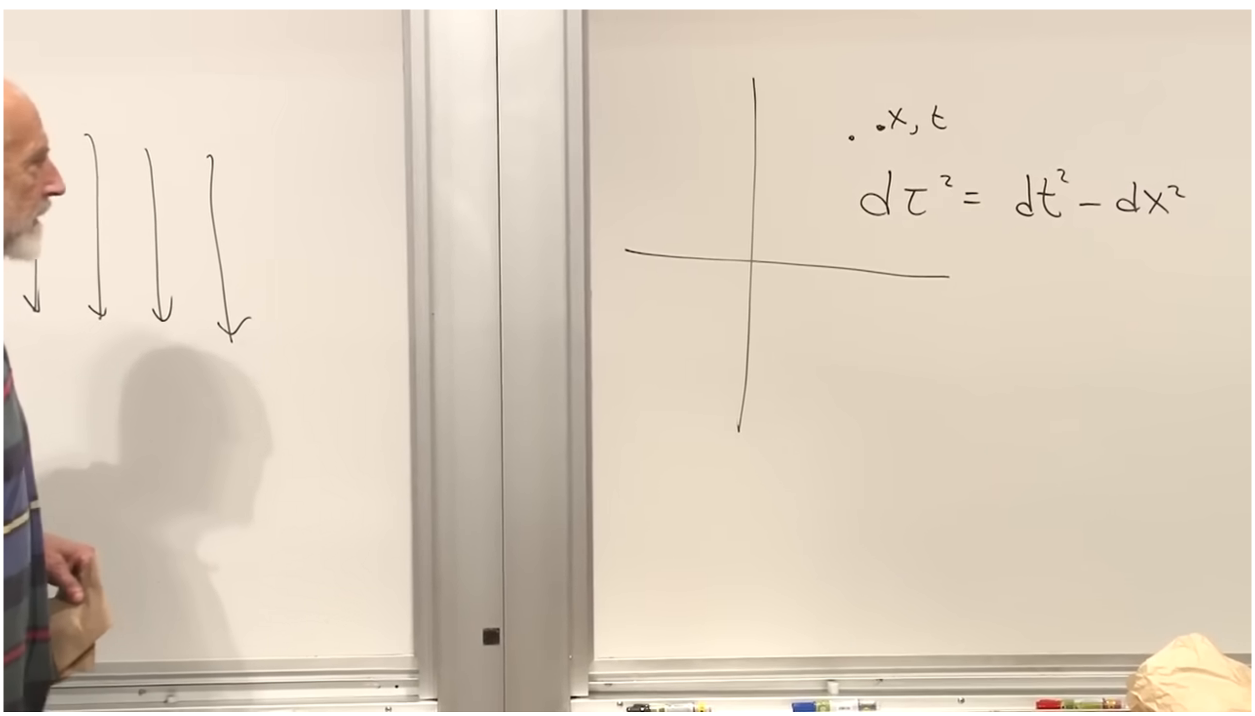

Figure 1.10: The proper-time interval that supplies the geometric bridge from special to general relativity.

Lecture frame: 00:51:10.

Einstein recognized that the problem of deciding whether a gravitational field is removable resembles a problem studied by Gauss for surfaces and by Riemann in arbitrary dimensions: given a local distance law, how can one decide whether the geometry is genuinely curved?

In Euclidean Cartesian coordinates, Pythagoras gives

$$ds^2 = (dx^1)^2 + (dx^2)^2 + \cdots = \delta_{mn}\, dx^m dx^n. \tag{1.13}$$

The index range is the dimension of the space. Nothing in the notation changes whether there are two, three, or twenty-six coordinates.

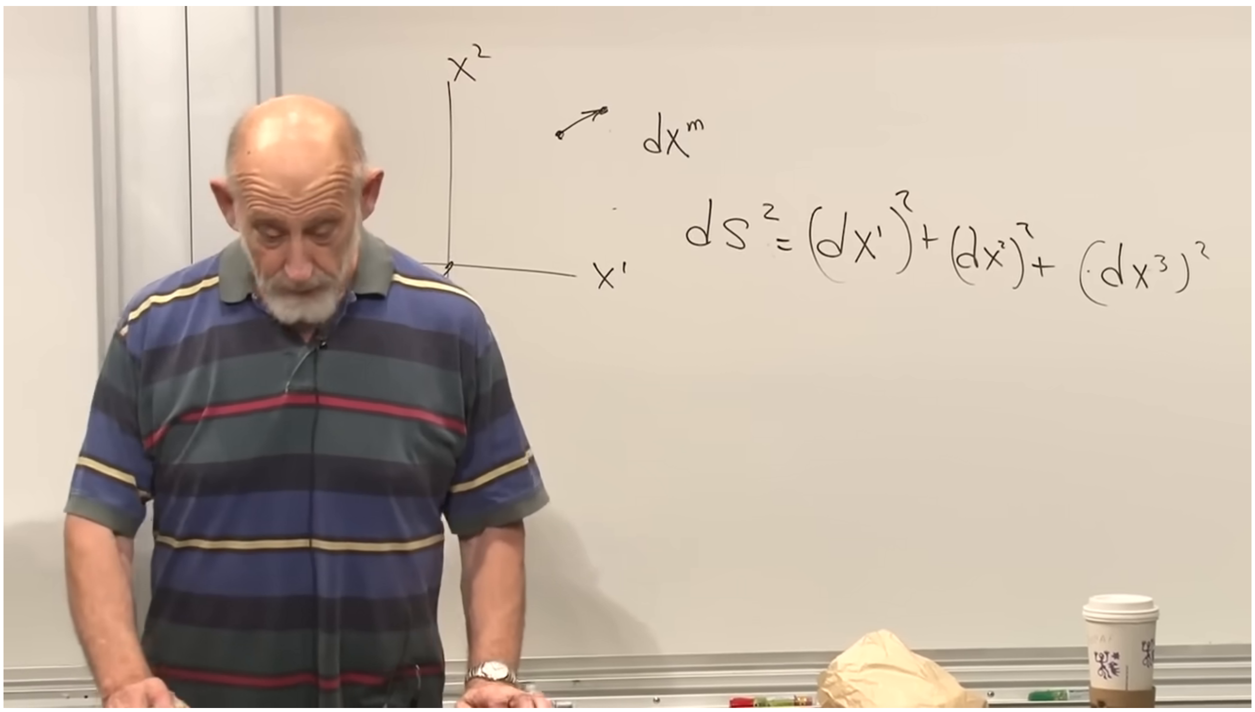

Figure 1.11: The Euclidean line element as a sum of squared coordinate increments.
Lecture frame: 00:55:00.

Place arbitrary coordinates x^m on an arbitrary surface. The distance between neighboring points is the quadratic form

$$ds^2 = g_{mn}(x)\, dx^m dx^n. \tag{1.14}$$

The functions $g_{mn}(x)$ are the components of the metric tensor. The same formula also applies to a flat surface described by curvilinear coordinates; nonconstant metric components do not by themselves prove intrinsic curvature.

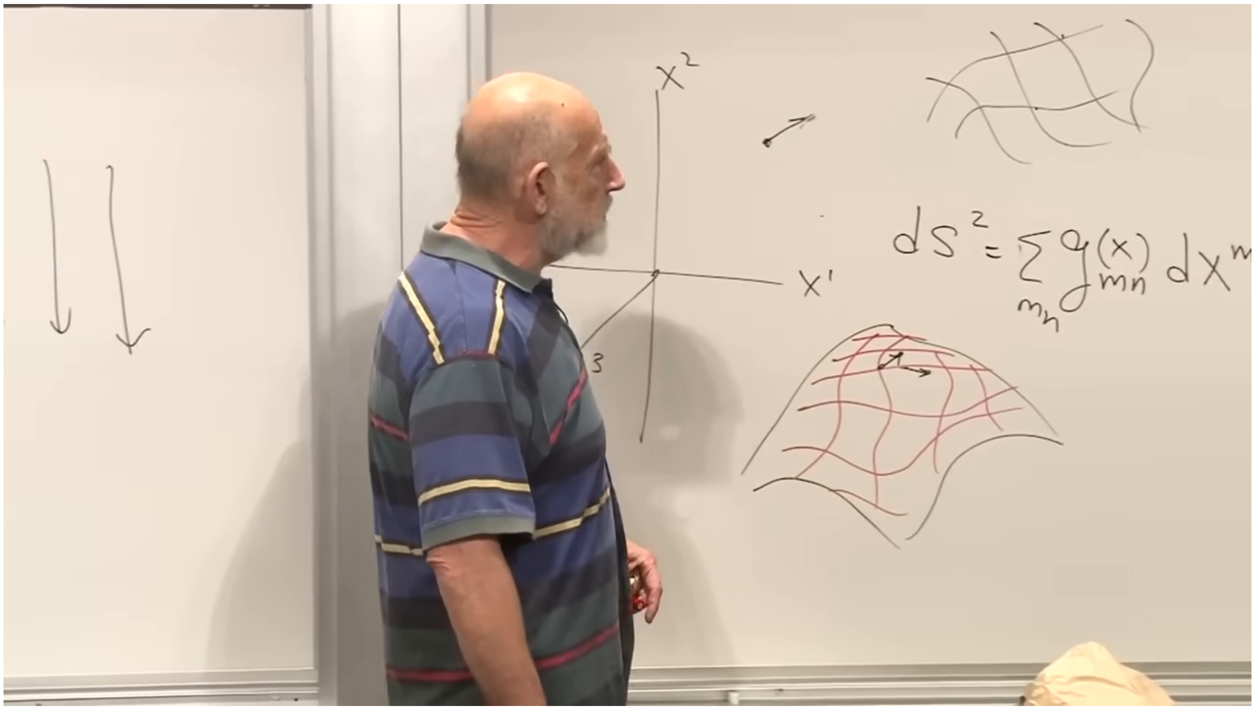

Figure 1.12: A general metric quadratic form beside a curved coordinate net.
Lecture frame: 00:58:20.

For a sphere of radius R, when θ is measured from a pole,

$$ds^2 = R^2 \left(d\theta^2 + \sin^2\theta \, d\phi^2 \right). \tag{1.15}$$

The same longitude increment covers a large distance near the equator and a small distance near a pole; that position dependence is exactly what the coefficient records.

Question from the audience. Ordinary latitude is measured from the equator, so should the coefficient be a cosine rather than a sine?

Response. Yes. If λ denotes latitude measured from the equator, then $ds^2 = R^2(d\lambda^2 + \cos^2\lambda \, d\phi^2)$. Equation (1.15) uses the polar angle $\theta = \pi/2 - \lambda$, measured from the north pole. [a]

[a] Lecture timestamp: 01:01:03.

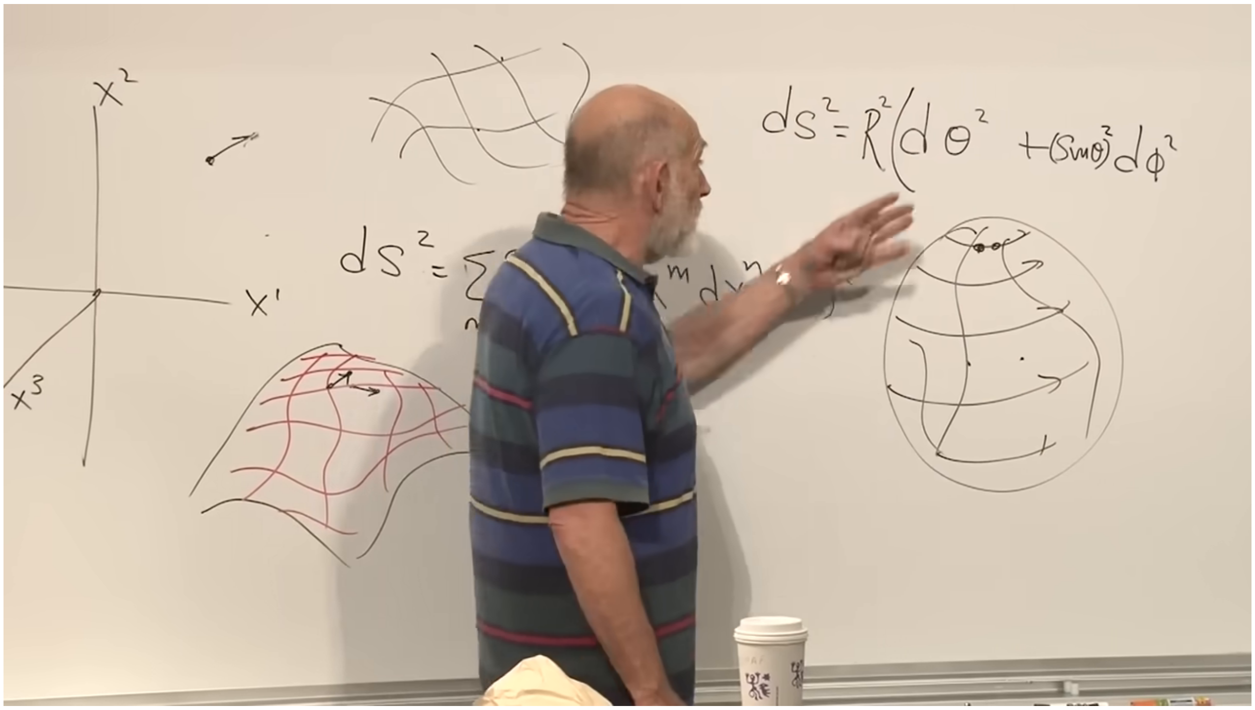

Figure 1.13: The sphere metric and the latitude–longitude geometry that fixes its position-dependent coefficient.

Lecture frame: 01:00:30.

More general coordinates can introduce $d\theta\, d\phi$ cross terms, but the local distance remains quadratic in the differentials. Cubic or linear terms do not belong to a Riemannian line element.

1.5 Local Distance and Intrinsic Curvature

Question from the audience. Does the metric formula also give the distance between two widely separated points?

Response. Not by direct substitution. It gives the distance between neighboring points. A macroscopic distance requires a path and an integral of ds; if distance means the shortest path, that path must itself be found from the geometry. Several locally shortest paths may even connect the same endpoints. [a]

[a] Lecture timestamp: 01:02:30.

On a bumpy surface, imagine pegs at two distant points and a taut string between them. The string may pass around one side of a hill or another, and moving a hill changes the minimizing route. Global distance depends on the geometry in between. Local distance is simpler: the metric supplies a small stick at every point, and the geometry is assembled from those infinitesimal sticks.

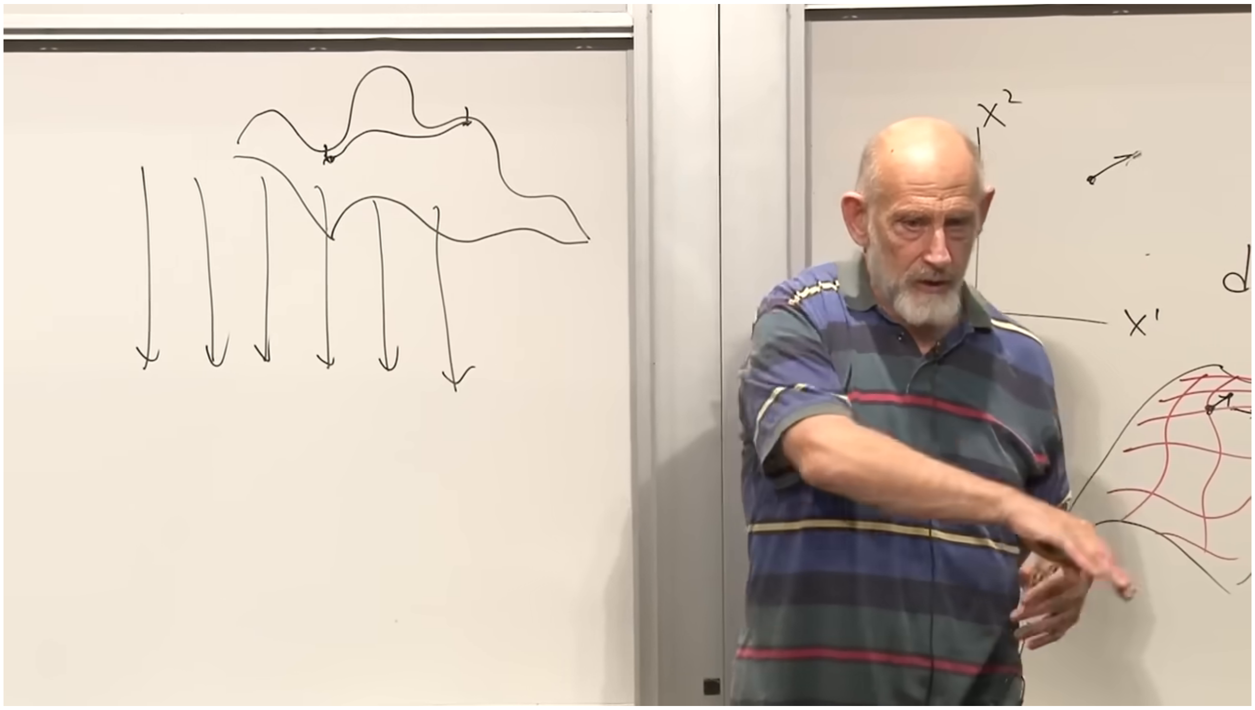

Figure 1.14: The global-distance problem: a taut route must negotiate the surface between its endpoints.

Lecture frame: 01:04:00.

If the local pieces can be laid onto a plane without stretching or tearing, the intrinsic geometry is flat. Pieces with the lengths of a triangulated sphere cannot all be laid flat while preserving their edges. The metric therefore poses a precise problem: given $g_{mn}(x)$, can one determine whether its infinitesimal pieces fit together as a flat space?

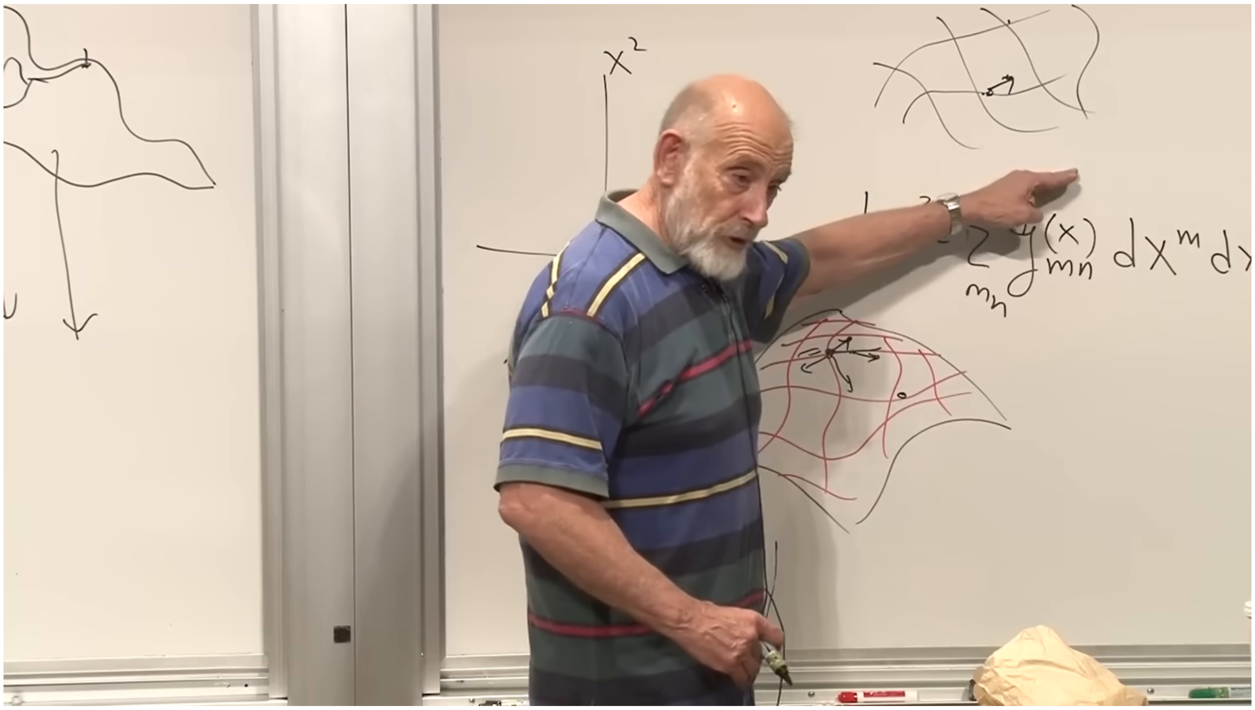

Figure 1.15: The metric and local coordinate patch interpreted as the infinitesimal pieces from which geometry is assembled.

Lecture frame: 01:05:20.

Question from the audience. Could one diagnose curvature by measuring the ratio of circumference to diameter?

Response. Yes, that is a useful two-dimensional test. On a curved surface the ratio can differ from π. For an infinitesimal circle the deviation enters at higher order, however, and the method does not generalize as cleanly as a curvature tensor. [a]

[a] Lecture timestamp: 01:07:26.

A region is flat if there exists a coordinate system in which

$$g_{mn} = \delta_{mn}, \qquad ds^2 = \delta_{mn}\, dx^m dx^n \tag{1.16}$$

throughout that region. A complicated $g_{mn}(x)$ may therefore describe either a curved space or merely curved coordinates on a flat space. The question is whether a coordinate transformation can remove all of the complication.

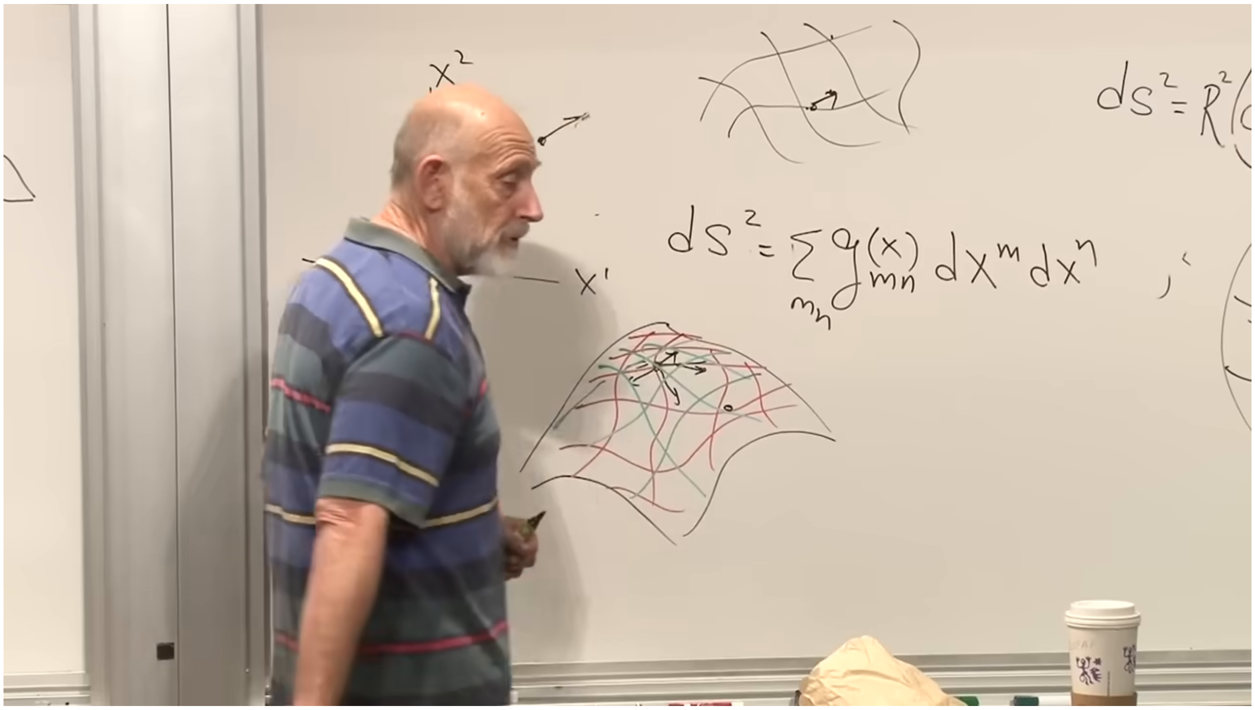

Figure 1.16: The metric problem stated on a local coordinate patch: can its curvilinear form be transformed to the flat one?

Lecture frame: 01:07:15.

Question from the audience. Does the spherical line element assume that each small interval is flat even though the sphere is not?

Response. To leading order, every smooth surface is approximated at a point by its tangent plane. Curvature appears in how neighboring tangent approximations fail to fit together, or in higher-order changes across a finite patch. Local Euclidean behavior and global curvature are therefore compatible. [a]

[a]Lecture timestamp: 01:10:01.

Question from the audience. Must the metric be continuous, and what happens at a crease or crevice?

Response. The discussion assumes coordinate maps and a metric smooth enough for the required derivatives. A sheet folded without stretching has the same intrinsic distances on both sides of the fold; its dramatic appearance is extrinsic. A genuinely singular or nonsmooth geometry requires separate care and is outside the smooth setup used here. [a]

[a]Lecture timestamp: 01:11:25.

This distinction is intrinsic geometry. Bending a page in three dimensional space changes its embedding but not distances measured within the page. A creature confined to the page, unable to look out into the embedding space, can still measure its intrinsic geometry.

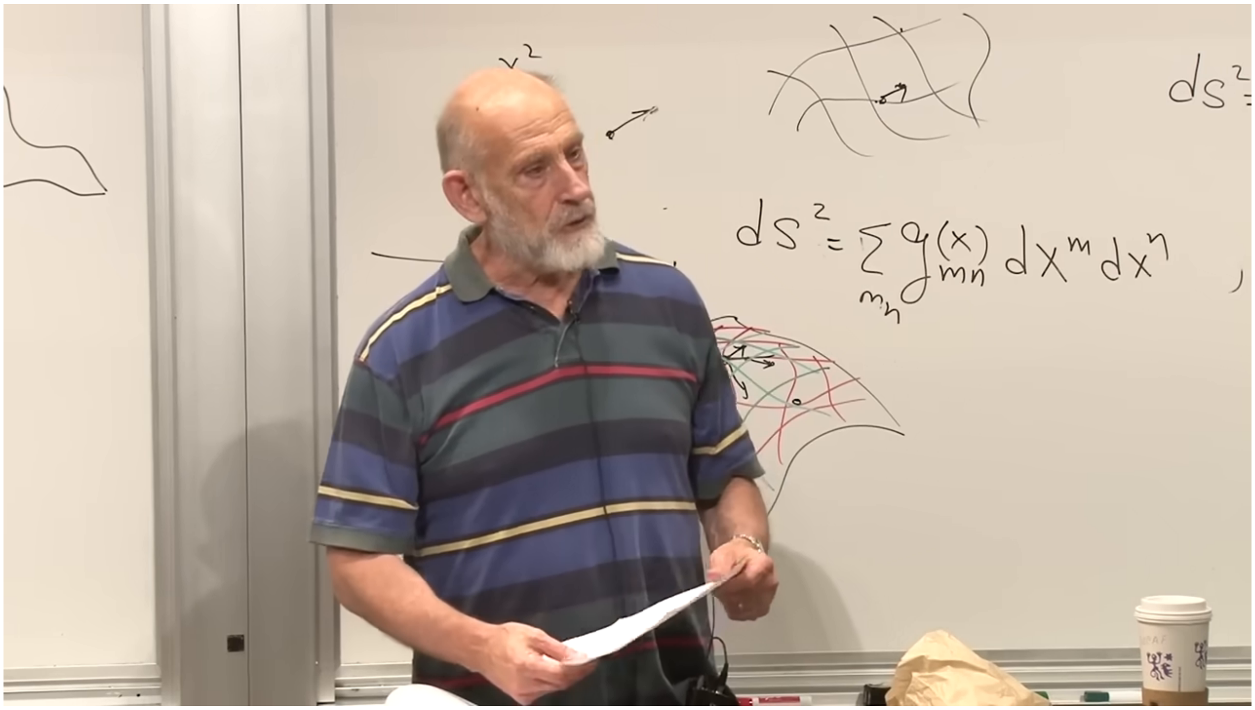

Figure 1.17: A folded sheet used to separate intrinsic distance from extrinsic appearance. *Lecture frame: 01:13:00.*

Question from the audience. Why not simply diagonalize the metric?

Response. At any one point a suitable coordinate choice can diagonalize the metric and normalize it to the local flat form. The obstruction is doing so throughout a region with one coordinate system. Curvature cannot be detected by inspecting a single point alone. [a]

[a]Lecture timestamp: 01:13:48.

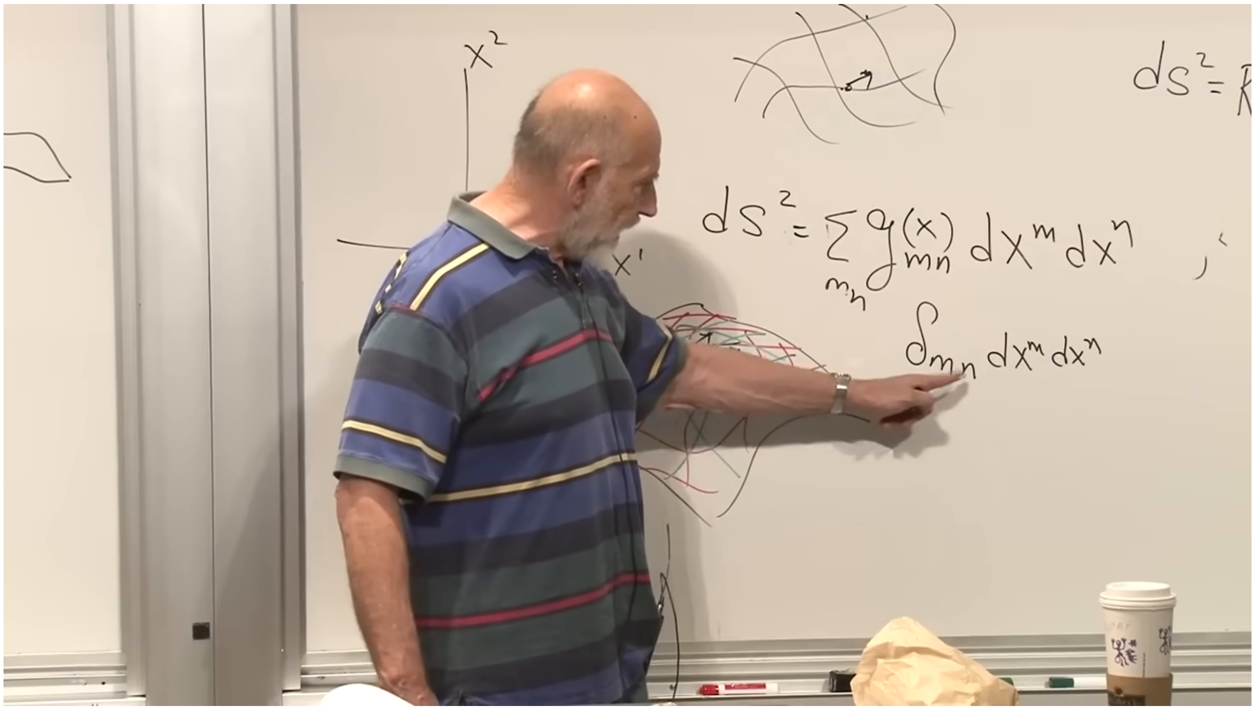

Figure 1.18: The flat Cartesian metric δ_{mn} written beneath the general local quadratic form.
Lecture frame: 01:16:00.

The parallel with gravity is now exact rather than suggestive. A coordinate force is removable just as a curvilinear description of flat space is removable. Tidal acceleration is measured in general relativity by curvature. In later notation, neighboring freely falling worldlines satisfy the geodesic-deviation equation

$$\frac{D^2\xi^\mu}{D\tau^2} = -R^\mu{}_{\nu\rho\sigma}u^\nu\xi^\rho u^\sigma. \tag{1.17}$$

Vanishing curvature means no invariant tidal field; nonzero curvature means that no coordinate choice can remove gravity throughout the region.

1.6 Two Coordinate Systems and One Vector

To decide whether a metric can be transformed to δ_{mn}, we must first know how geometric objects transform. Let x^m and y^m be two smooth coordinate systems on the same region:

$$x^m = x^m(y), \qquad y^m = y^m(x). \tag{1.18}$$

The symbols x and y label complete coordinate systems, not two particular spatial directions.

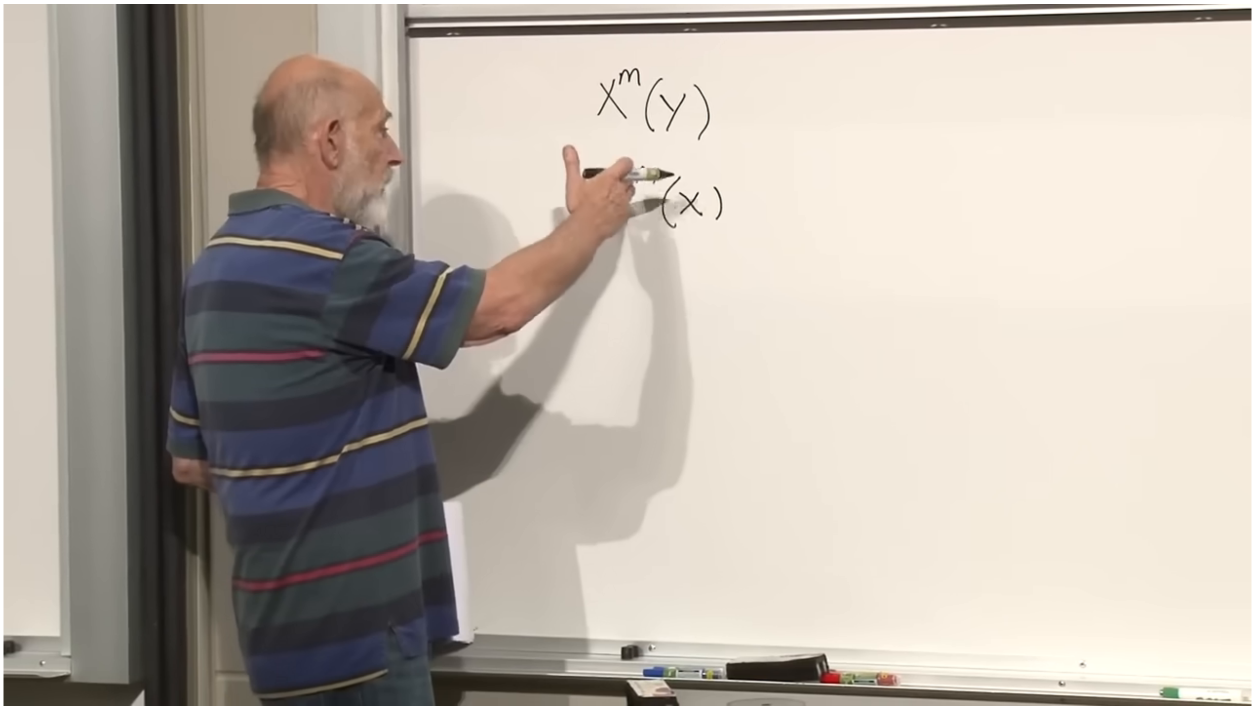

Figure 1.19: The mutually related coordinate maps $x^m(y)$ and $y^m(x)$.
Lecture frame: 01:20:35.

For neighboring points, the chain rule gives

$$dy^m = \frac{\partial y^m}{\partial x^p}\, dx^p. \tag{1.19}$$

The repeated index p is summed. The displacement itself is one geometric vector; dx^p and dy^m are its components in two coordinate systems.

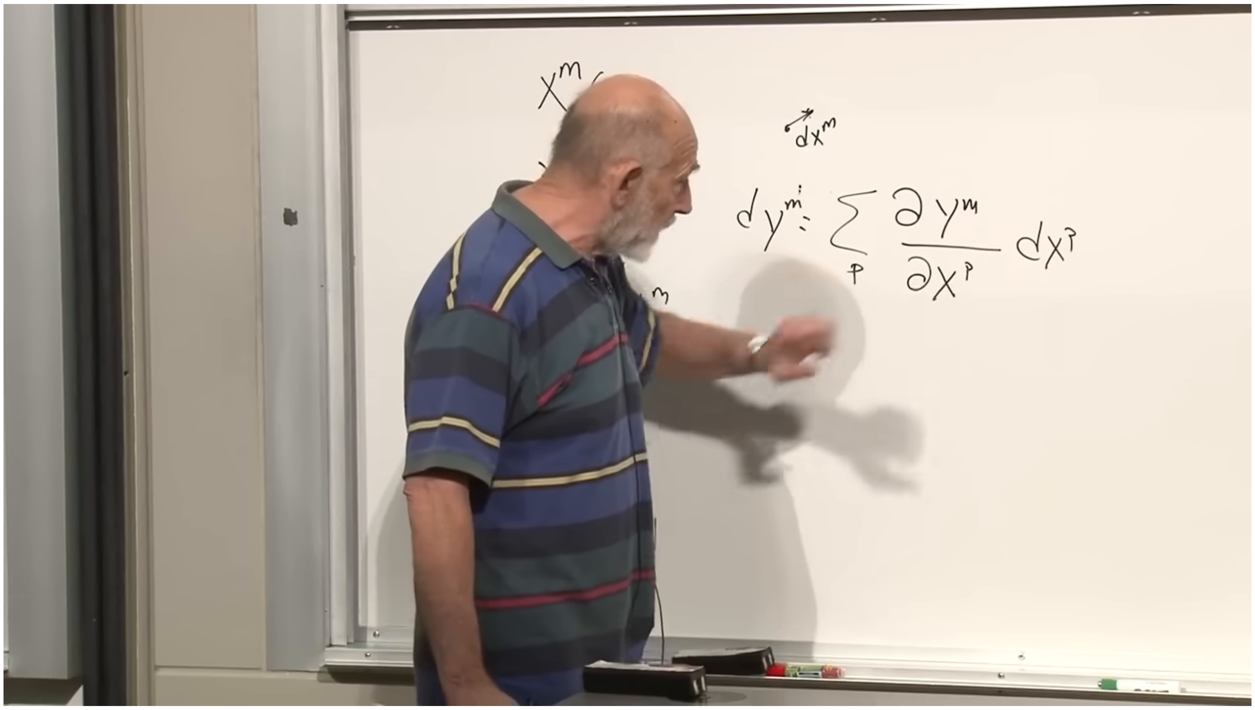

Figure 1.20: The differential coordinate transformation obtained from the chain rule.
Lecture frame: 01:22:30.

Question from the audience. On a curvilinear grid, what do the coordinate lines represent?

Response. In two dimensions, a line is obtained by holding one coordinate fixed while varying the other. In higher dimensions the corresponding constant-coordinate sets are hypersurfaces. Their local intersections define the coordinate directions used to resolve vector components. [a]

[a] Lecture timestamp: 01:25:20.

A contravariant vector is defined by the same transformation law as a coordinate displacement:

$$(V')^m(y) = \frac{\partial y^m}{\partial x^p} V^p(x). \tag{1.20}$$

Velocity along a surface is an example. The vector does not change when coordinates change; its components do.

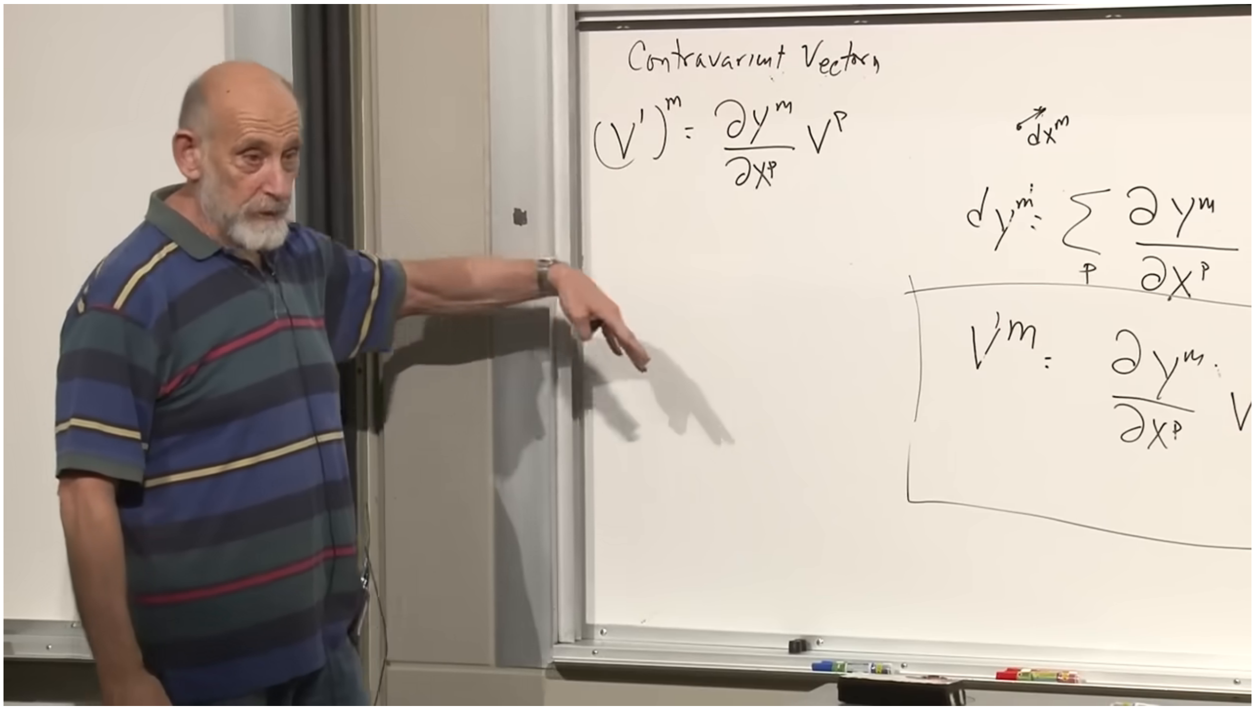

Figure 1.21: The boxed transformation law for a contravariant vector.

Lecture frame: 01:29:30.

Einstein's summation convention suppresses the summation sign whenever an index is repeated once upstairs and once downstairs. The convention made formulas easier to print, but more importantly it makes the contraction pattern visible.

Question from the audience. Are special assumptions being made about which coordinate systems are allowed?

Response. Only that the two systems are functionally related and smooth enough for the displayed derivatives and inverse map to exist in the region under discussion. No linearity or orthogonality is assumed. [a]

[a] Lecture timestamp: 01:28:12.

1.7 Covectors and the Chain Rule

Let $S(x)$ be a scalar: it has one well-defined value at each point, independent of the coordinate labels. Temperature and a scalar field are examples. Its gradient has components

$$W_p = \frac{\partial S}{\partial x^p}. \tag{1.21}$$

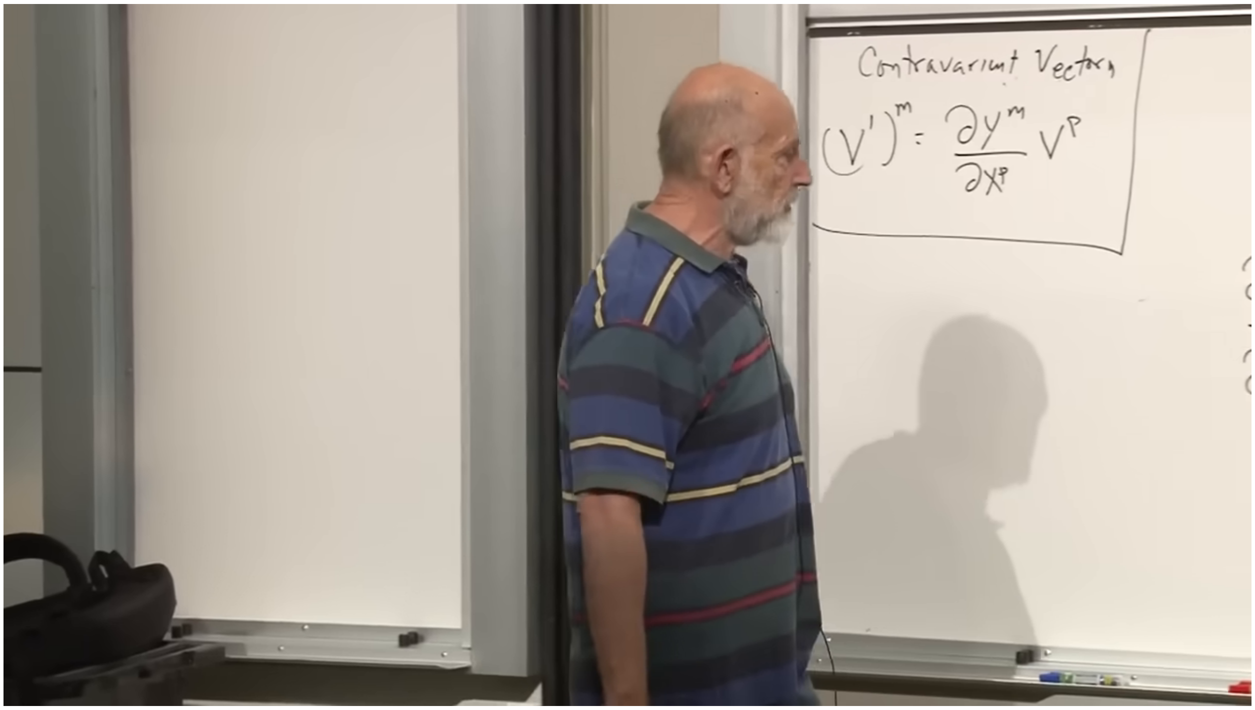

Figure 1.22: A scalar gradient introduced beside the contravariant transformation law.
Lecture frame: 01:31:40.

Applying the chain rule in the y coordinates,

$$W'_m = \frac{\partial S}{\partial y^m} = \frac{\partial x^p}{\partial y^m}\frac{\partial S}{\partial x^p} = \frac{\partial x^p}{\partial y^m}\, W_p. \tag{1.22}$$

This is the transformation law of a covariant vector, or covector. The Jacobian is the inverse of the one in (1.20); lower and upper indices therefore encode different transformation behavior.

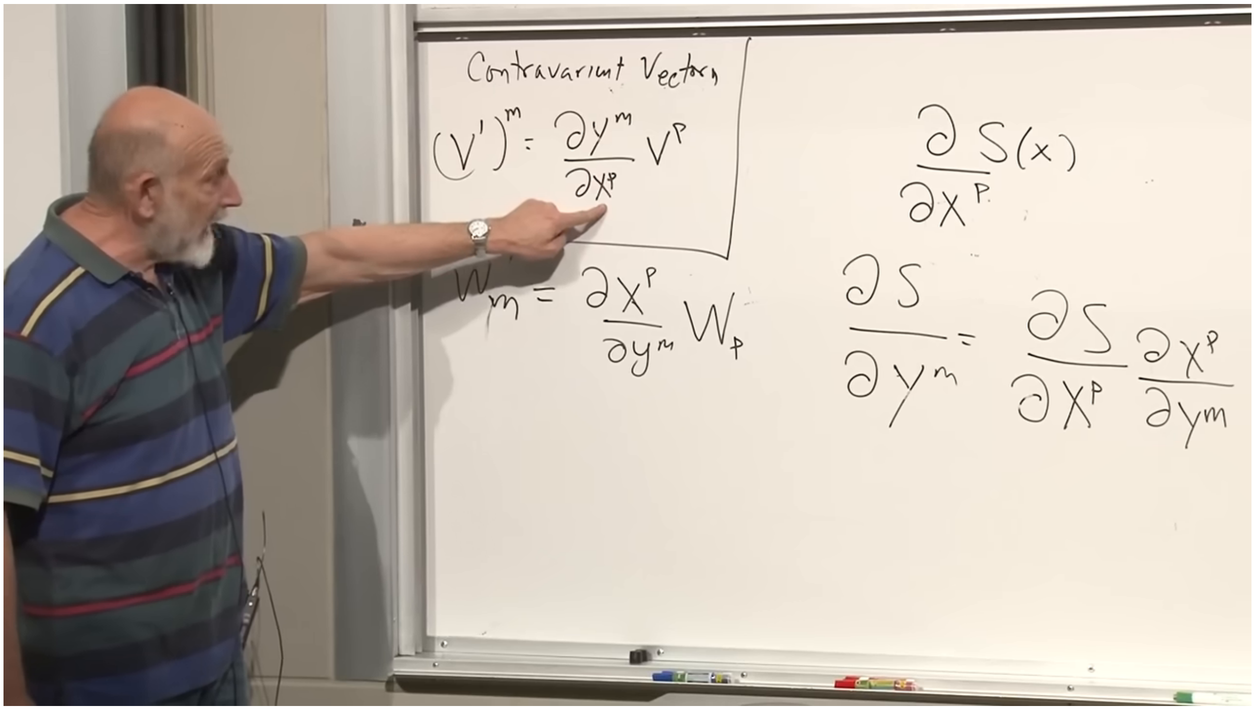

Figure 1.23: Contravariant and covariant transformation laws displayed together.
Lecture frame: 01:34:40.

Question from the audience. What is the connection between covariant and contravariant vectors?

Response. At this stage they are defined as different kinds of objects: displacements and velocities transform one way, gradients the inverse way. The metric will later provide a one-to-one map that lowers or raises an index. That relationship should not be assumed before the metric is introduced. [a]

[a]Lecture timestamp: 01:37:43.

Question from the audience. Did lowering an index previously change the sign of the time component?

Response. That sign came from the Lorentzian metric. The present derivation is first being done in ordinary Euclidean geometry. Raising or lowering always uses the relevant metric; a time-component sign appears only when that metric has Lorentzian signature. [a]

[a]Lecture timestamp: 01:38:29.

Question from the audience. Should the two transformation laws already be viewed as two component descriptions of the same vector?

Response. Not yet. Once the metric is available, a contravariant vector V^m can be converted to a covector $V_m = g_{mn}V^n$. Until that map has been established, upper- and lower-index objects are logically distinct. [a]

[a]Lecture timestamp: 01:39:00.

These formulas are independent of dimension. The index range changes, but the transformation laws do not.

1.8 Tensors with More Than One Index

Take two contravariant vectors V^m and U^n. Their product

$$T^{mn} = V^m U^n \tag{1.23}$$

is a rank-two contravariant tensor. In three dimensions it has nine components; in four dimensions it has sixteen. The order of the indices matters unless a separate symmetry is given: T^{mn} need not equal T^{nm}.

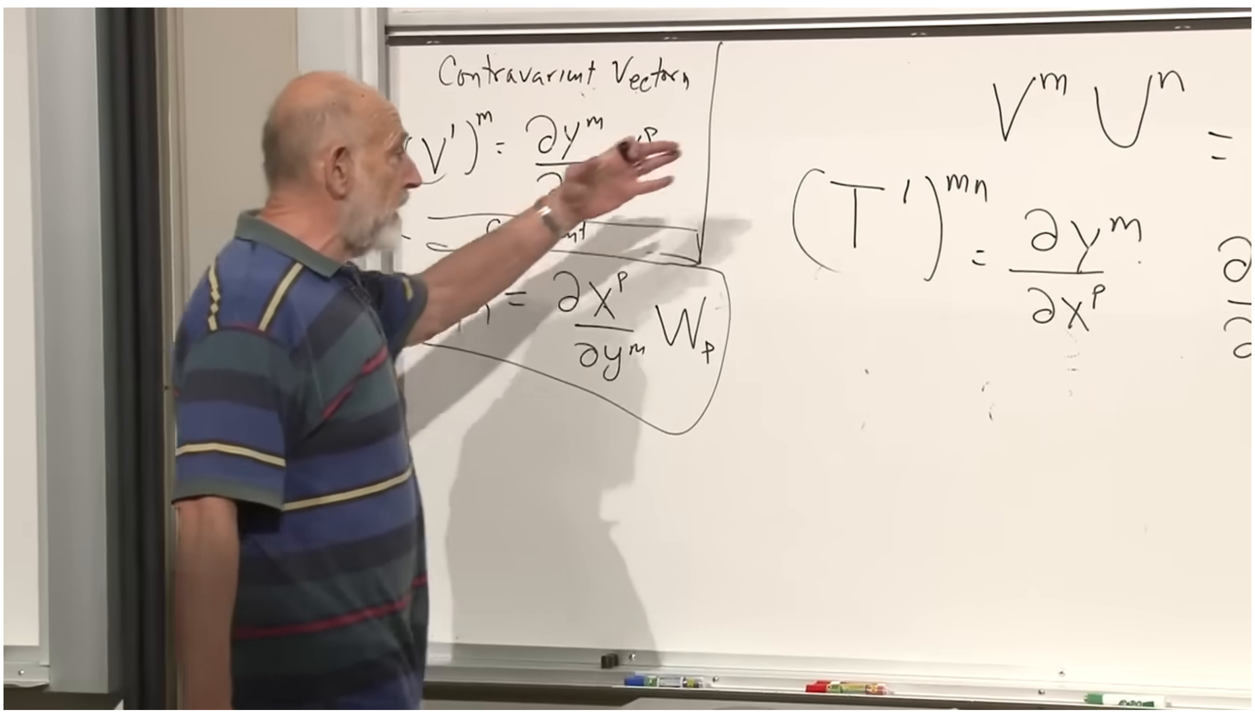

Figure 1.24: A rank-two tensor built as the product of two contravariant vectors.
Lecture frame: 01:43:40.

Each upper index contributes one forward Jacobian:

$$(T')^{mn} = \frac{\partial y^m}{\partial x^p}\frac{\partial y^n}{\partial x^q}T^{pq}. \tag{1.24}$$

The product construction motivates the rule, but the transformation law is the definition of a tensor. A rank-three contravariant tensor would acquire three such Jacobian factors.

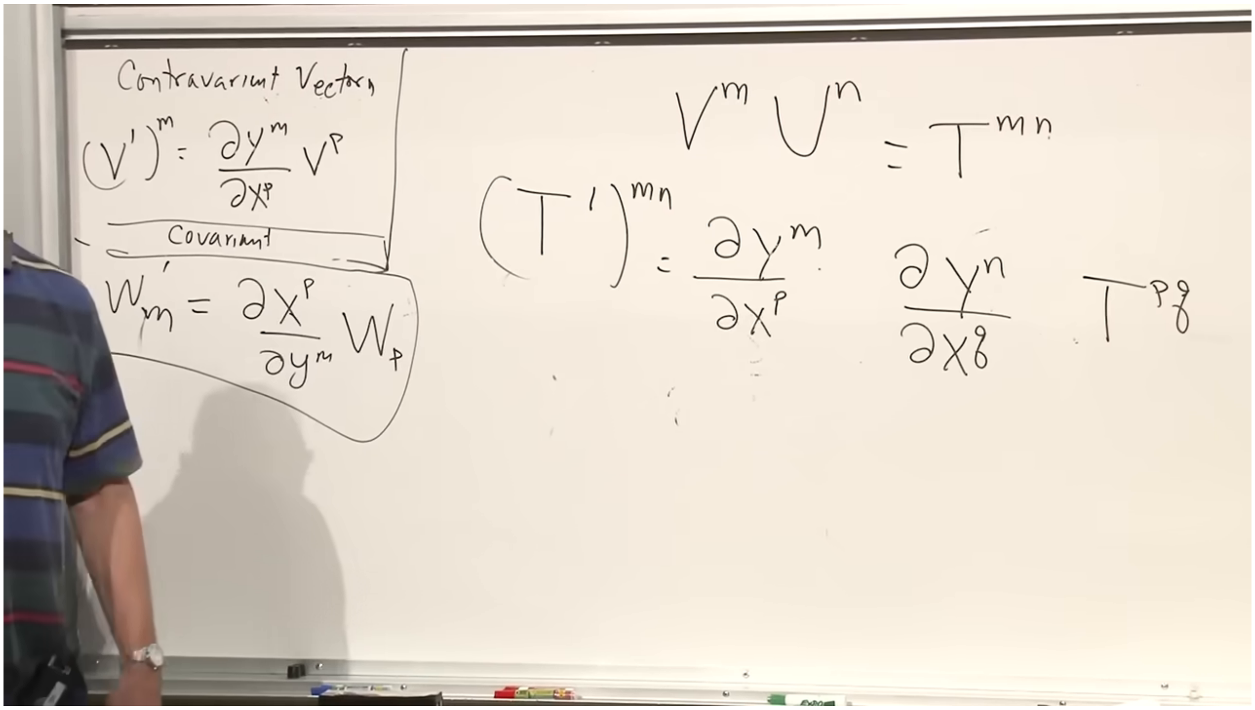

Figure 1.25: The two Jacobian factors in the transformation of T^{mn}.
Lecture frame: 01:44:20.

Likewise, each lower index contributes an inverse Jacobian:

$$T'_{mn} = \frac{\partial x^p}{\partial y^m}\frac{\partial x^q}{\partial y^n}T_{pq}. \tag{1.25}$$

A mixed tensor uses one factor of each kind.

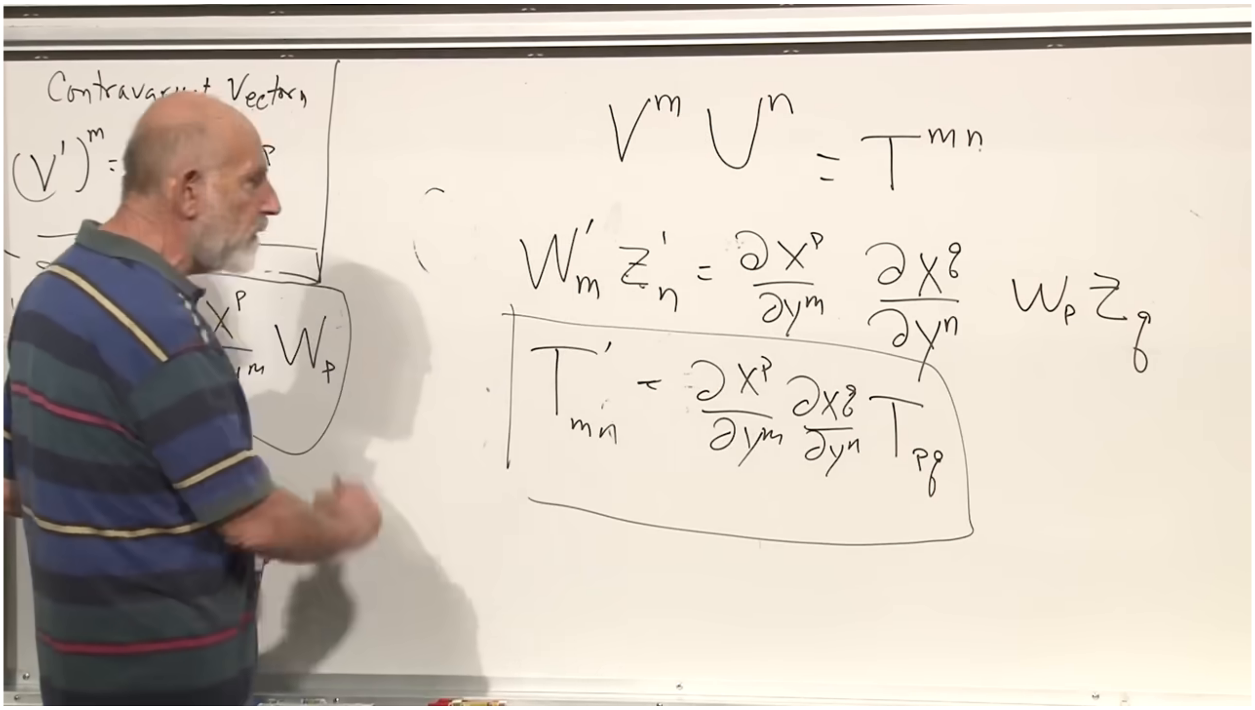

Figure 1.26: The completed transformation law for a tensor with two covariant indices.
Lecture frame: 01:47:30.

The lecture ends just as this machinery returns to the metric. Since the interval is a scalar, it must have the same value in either coordinate system:

$$ds^2 = g_{mn}(x)\, dx^m dx^n, \tag{1.26}$$

$$dx^m = \frac{\partial x^m}{\partial y^p}\, dy^p. \tag{1.27}$$

Substitution completes the line that the board leaves unfinished:

$$g'_{pq}(y) = \frac{\partial x^m}{\partial y^p} \frac{\partial x^n}{\partial y^q} g_{mn}(x(y)). \tag{1.28}$$

Thus g_{mn} is a rank-two covariant tensor.

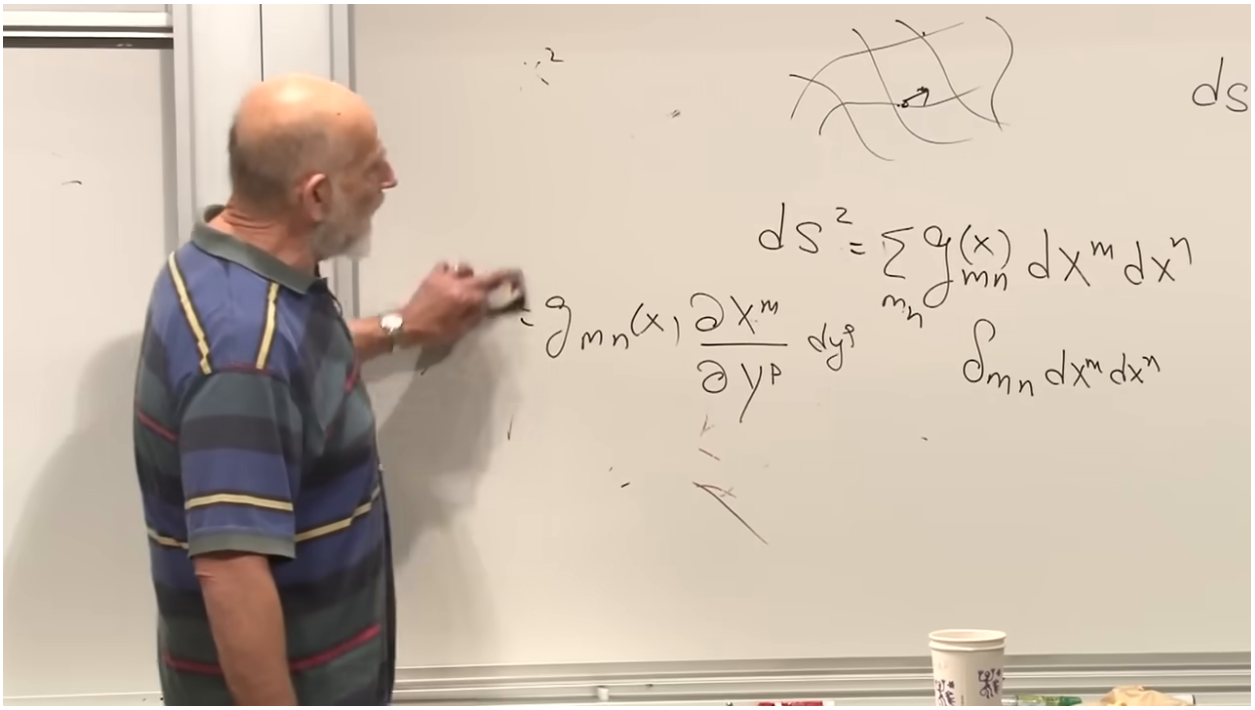

Figure 1.27: The metric transformation begun from invariance of the line element at the close of the lecture.

Lecture frame: 01:48:40.

The opening physical question has now become a mathematical one. Given $g_{mn}(x)$, does there exist a coordinate transformation for which $g'_{mn} = \delta_{mn}$ throughout the region? If so, the apparent complexity is only coordinate complexity. If not, the obstruction is curvature, and in spacetime that obstruction is the invariant content of gravity.